Appendix A: Literature Evidence Matrices

A.1 RVI-01: Reproducible Benchmarking — Literature Evidence Matrix

Paper	Authors & Venue	DOI	Research Problem	Core Methodology	Robot Platform	Dataset/ Environment	Evaluation Metrics	Main Findings	Limitations	Relevance to ARC-RVI
VPR-Bench 2	Zaffar et al., 2021, IJCV	10.1007/ S11263-021-01469-5	Fragmentation and lack of standardization in visual place recognition evaluation	Open-source framework integrating 12 datasets, 10 VPR techniques, multiple metrics; uses precision robot arm with LED arrays for quantifiable viewpoint/ illumination variation	Precision robot arm with LED illumination array	12 integrated datasets including QUT multi-lane, MIT multi-illumination	Recall@N, precision-recall curves, viewpoint/ illumination invariance scores	Numerical and visual analysis of technique invariance limits and performance trade-offs	Limited to place recognition; does not address inspection-specific quality metrics	High: Demonstrates controlled environmental variation and quantifiable benchmarking
RobotArena∞ 3	Jangir et al., 2025	10.48550/ arxiv.2510.23571	Labor-intensive, non-scalable real-world testing and synthetic simulation limitations	Automatically converts robot demonstration videos into simulated digital twins using VLMs, 2D-to-3D generative modeling, differentiable rendering	Robot datasets converted to simulation	Large-scale simulated environments from robot datasets	VLM-guided scoring, human preference judgments, robustness under perturbations	Scalable benchmarking via real-to-sim translation; automated scoring and systematic perturbation testing	Simulation-only; no physical robot validation of converted environments	Medium: Demonstrates scalable benchmarking but lacks physical validation
ROEP 4	Han & Choi, 2026, Sensors	10.3390/s26154757	Success rate alone obscures runtime fidelity, observation degradation, and recovery semantics	Deployment-oriented protocol converting rollout outcomes into claim-level evidence; evaluates clean-condition evaluability, runtime fidelity, visual perturbations, repeated-run variability	Simulation (LIBERO Goal and Object tasks)	Eight LIBERO Goal and Object tasks	Clean-condition success, runtime fidelity, visual perturbation robustness, timeout failures, recovery support	Reveals hidden vulnerabilities in VLA policies; systematic evaluation of deployment-relevant factors	Simulation-only; no physical robot validation	High: Emphasizes deployment-oriented evaluation and controlled perturbations
Grasping Benchmark 5	Rameshbabu et al., 2025, arXiv	10.48550/arxiv.2503.11163	Lack of standardized benchmarking for vision-based grasping algorithms	Evaluates four algorithms (GG-CNN, ResNet, Top Surface, Mask-based) across 5,040 experiments with varying lighting, backgrounds, cameras, grippers, manipulators	Franka Panda, UR5e with Intel RealSense D435, ZED 2i cameras	Ten YCB objects under varying conditions	Grasp success rate, execution time	Systematic comparison demonstrates importance of standardized protocols	Limited to grasping; does not address inspection or documentation quality	Medium: Demonstrates standardized benchmarking but limited to manipulation
AgroBench 6	Zarrouk et al., 2026, MethodsX	10.1016/ j.mex.2026.104068	Clean-set metrics conceal performance degradation under realistic field perturbations	Agriculture-calibrated corruption benchmark with explicit corruption families, parameter grids, global modulation; Agro-Robustness Index (ARI) summarizes corrupted performance	Agricultural object detection (YOLOv8n)	Agriculture-calibrated corruption protocol	ARI (normalized retention score relative to clean reference)	Reproducible evaluation method quantifying performance retention under domain-grounded corruptions	Limited to agricultural object detection; no robot platform validation	Medium: Demonstrates reproducible corruption-based evaluation
RoboBEV 7	Xie et al., 2023, arXiv	10.48550/ arXiv.2304.06719	Unexplored out-of-distribution robustness of camera-based BEV 3D perception under natural corruptions	Benchmark suite with eight corruptions across three severities; evaluates 26 BEV perception models using nuScenes-C	Autonomous driving perception (26 BEV models)	nuScenes-C (corrupted validation set)	mAP, NDS under corruptions	Systematic evaluation reveals robustness factors (pre-training, depth-free transformations, temporal fusion)	Limited to autonomous driving; no robot platform validation	Low: Domain-specific (autonomous driving); limited relevance to inspection
Noisy-Replica 8	Xu et al., 2024	10.48550/ arxiv.2406.16850	SLAM robustness benchmarking for embodied agents in noisy environments	Customizable perturbation taxonomy (sensor pose errors, RGB-D corruptions, motion deviations, de-synchronization); noisy data synthesis	Mobile robots and embodied agents (simulation)	Noisy-Replica benchmark (1,000 video sequences from Replica dataset)	Trajectory error, robustness under isolated/ combined perturbations	Systematic evaluation of neural and non-neural SLAM models under realistic perturbations	Simulation-only; no physical robot validation	Medium: Demonstrates systematic perturbation testing but lacks physical validation

Paper	Authors & Venue	DOI	Research Problem	Core Methodology	Robot Platform	Dataset/ Environment	Evaluation Metrics	Main Findings	Limitations	Relevance to ARC-RVI
				pipeline						
RLBench/ Colosseum 9	Matar et al., 2026	10.1109/ aeect69724.2026.11657856	Robustness under test-time shift for camera-guided industrial manipulation	Compares three training regimes (baseline, targeted domain randomization, adaptive hard-example sampling)	Fixed-camera Franka Panda workcell (simulation)	RLBench and Colosseum (four tasks)	Success rate under test-time lighting, background, table, object appearance shifts	Adaptive hard-example sampling improves robustness under test-time perturbations	Simulation-only; no physical robot validation	Medium: Demonstrates robustness evaluation but lacks physical validation
RobustSpring 10	Schmalfuss et al., 2025	10.48550/ arxiv.2505.09368	Unquantified resilience of optical flow, scene flow, stereo vision to real-world image corruptions	Dataset applying 20 diverse corruptions in time-, stereo-, depth-consistent manner to Spring dataset; ground-truth-free Lipschitz continuity metric	Dense matching algorithms (simulation + KITTI transfer)	RobustSpring (20,000 corrupted stereo frame pairs)	Lipschitz continuity, average/ median/ Schulze rankings	Systematic evaluation reveals corruption robustness; transfer to KITTI validates findings	Limited to dense matching; no robot platform validation	Low: Domain-specific (dense matching); limited relevance to inspection

Summary: Existing benchmarks emphasize robustness to visual corruptions and perturbations but neglect inspection-specific requirements like acquisition geometry repeatability, documentation quality, and humanoid platform validation. Most studies rely on simulation or domain-specific datasets (autonomous driving, agriculture) rather than general-purpose inspection tasks.

A.2 RVI-02: Target Identification and Localization — Literature Evidence Matrix

Paper	Authors & Venue	DOI	Research Problem	Core Methodology	Robot Platform	Dataset/ Environment	Evaluation Metrics	Main Findings	Limitations	Relevance to ARC-RVI
Active 6D Pose 6	Sock et al., 2019	—	6D multi-object pose estimation in cluttered scenarios	Deep reinforcement learning for active viewpoint selection	Not specified	Cluttered multi-object scenes	Pose estimation accuracy	Active perception improves pose estimation in clutter	Limited details on robot platform and physical validation	Medium: Demonstrates active perception for pose estimation
ActivePose 8	Sheng et al., 2025	10.48550/ arxiv.2509.11364	Active 6D object pose estimation and tracking for manipulation	Combines pose tracking with active viewpoint selection	Not specified	Manipulation tasks	Pose estimation accuracy, tracking performance	Active viewpoint selection improves pose estimation and tracking	Limited details on physical validation	Medium: Demonstrates active pose estimation for manipulation
REACT 9	Nguyen et al., 2025, arXiv	10.48550/ arxiv.2503.03412	Real-time efficient attribute clustering and transfer for updatable 3D scene graphs	Attribute clustering and transfer for 3D scene understanding	Not specified	3D scene graph datasets	Scene graph accuracy, update efficiency	Efficient real-time 3D scene graph updates	Limited details on robot platform and physical validation	Low: Focuses on scene understanding, not inspection
Smart Explorer 10	2022	10.1109/ iros47612.2022.9981813	Recognizing objects in dense clutter via interactive exploration	Interactive exploration to resolve ambiguities	Not specified	Dense clutter scenarios	Object recognition accuracy	Interactive exploration improves recognition in clutter	Limited details on physical validation	Medium: Demonstrates interactive exploration for object recognition
SE(3)-PoseFlow 11	Yu-feng et al., 2025	10.48550/ arxiv.2511.01501	Estimating 6D pose distributions for uncertainty-aware manipulation	Probabilistic pose estimation with uncertainty quantification	Not specified	Manipulation tasks	Pose estimation accuracy, uncertainty calibration	Uncertainty-aware pose estimation improves manipulation reliability	Limited details on physical validation	Medium: Demonstrates uncertainty quantification for pose estimation
LidaRefer 12	Baek et al., 2024	10.48550/ arxiv.2411.04351	Outdoor 3D visual grounding for autonomous driving	Transformer-based 3D visual grounding	Autonomous driving platform	Outdoor driving datasets	Grounding accuracy	Effective 3D visual grounding for autonomous driving	Domain-specific (autonomous driving); limited relevance to inspection	Low: Domain-specific; limited relevance to inspection
Active Perception 13	Safronov et al., 2021	—	Ambiguous object classification	Active perception to resolve classification ambiguities	Not specified	Ambiguous object scenarios	Classification accuracy	Active perception improves classification of ambiguous objects	Limited details on physical validation	Medium: Demonstrates active perception for classification
Localization Uncertainty 14	2023	10.48550/ arxiv.2306.08981	Overcoming limitations of localization uncertainty	Efficient and exact non-linear post-processing and calibration	Not specified	Localization tasks	Localization accuracy, uncertainty calibration	Improved localization accuracy and uncertainty calibration	Limited details on physical validation	Medium: Demonstrates uncertainty calibration for localization
RoboEye 15	Xingwu et al., 2025	10.48550/ arxiv.2509.14966	Enhancing 2D robotic object identification with	Combines 2D detection with 3D geometric keypoint	Not specified	Object identification tasks	Identification accuracy	Improved object identification via 3D geometric	Limited details on physical validation	Medium: Demonstrates geometric keypoint matching for

Paper	Authors & Venue	DOI	Research Problem	Core Methodology	Robot Platform	Dataset/ Environment	Evaluation Metrics	Main Findings	Limitations	Relevance to ARC-RVI
			selective 3D geometric keypoint matching	matching				keypoint matching		identification
Fusion++ 16	McCormac et al., 2018, 3DV	10.1109/3DV.2018.00015	Volumetric object-level SLAM	Object-level SLAM with volumetric representation	Not specified	Indoor SLAM scenarios	SLAM accuracy, object-level mapping	Effective object-level SLAM with volumetric representation	Limited details on physical validation	Low: Focuses on SLAM, not inspection

Summary: Target identification and localization research emphasizes manipulation tasks and pose estimation accuracy, with limited attention to multi-instance identification in cluttered inspection scenarios, geometric measurement precision, or repeatability across sessions. Most methods lack physical robot validation, particularly on humanoid platforms.

A.3 RVI-03: Viewpoint and Motion Planning — Literature Evidence Matrix

Paper	Authors & Venue	DOI	Research Problem	Core Methodology	Robot Platform	Dataset/ Environment	Evaluation Metrics	Main Findings	Limitations	Relevance to ARC-RVI
Human-Inspired Inspection 1	Zou et al., 2026, Biomimetics	10.3390/ biomimetics11050296	Reliable image acquisition in complex geometries with severe occlusions	Hierarchical trajectory optimization inspired by human visual inspection; multi-constraint preprocessing for occlusion-aware viewpoint generation; Pose Reachability Augmented GAN for feasible imaging postures	Industrial robotic platform with 2D/3D vision	Simulation and real-world experiments	Coverage completeness, motion efficiency, kinematic feasibility	Balances global coverage and local motion efficiency; learns feasible postures under kinematic constraints	Limited details on humanoid platform applicability	High: Addresses occlusion-aware planning and kinematic feasibility
Manipulator-Aware Planning 2	Jafari-Tabrizi et al., 2026	10.1109/ aim65483.2026.11658262	Automated vision-based inspection of free-form surfaces with kinematic and collision constraints	Decomposes surfaces into geometric primitives; formulates scan path selection as GTSP; evaluates paths in joint-space with IK, collision avoidance, manipulability	Manipulator (simulation and real robot)	Representative 3D models	Coverage completeness, execution time, kinematic feasibility	Industry-oriented framework balances coverage and kinematic constraints	Limited to manipulators; no humanoid validation	High: Demonstrates joint-space planning with kinematic constraints
Visual Quality Inspection 3	Staderini et al., 2024, IROS	10.1109/ iros58592.2024.10802791	Integrating computer vision and robotics for visual quality inspection planning	Model-based framework for optimal inspection pose generation considering object geometry and robotic kinematics	Robotic setup (simulation and experiments)	Diverse geometries and materials	Coverage completeness, number of inspection poses	Complete coverage with minimal inspection poses	Limited details on humanoid platform applicability	High: Demonstrates model-based inspection planning
Cylindrical Cavity Inspection 4	Wang et al., 2026	10.48550/ arxiv.2608.16442	Viewpoint planning for observing bottom of cylindrical cavities	Formulates observation region with axis-alignment, lateral-offset, axial-standoff constraints; feasibility-first joint-space optimizer; collision-aware motion planning	6-DoF AgileX Piper manipulator with Intel RealSense D435i (tabletop and Unitree A2 quadruped-mounted)	Controlled simulations (Isaac Sim, 100 target configurations) and physical demonstrations	Kinematic feasibility, collision avoidance, visibility verification	Observation-constrained joint-space optimization enables reliable cavity inspection	Limited to manipulator on quadruped; no bipedal humanoid validation	High: Demonstrates joint-space optimization and physical validation on quadruped
Cluttered Pipe Networks 5	Bettles et al., 2026	10.48550/ arxiv.2603.16471	Autonomous inspection of unknown, confined, cluttered pipe networks	Coverage-first NBV planner combining visual exploration and coverage information gains; stochastic vector-field inequalities	UF7 manipulator on AgileX Tracer mobile platform	Confined space environment (1.5m cube with varying pipe configurations)	Coverage completeness, collision avoidance, execution time	Coverage-first planning with probabilistic collision avoidance enables reliable pipe network inspection	Limited to mobile manipulator; no humanoid validation	High: Demonstrates coverage-first planning with collision avoidance

Paper	Authors & Venue	DOI	Research Problem	Core Methodology	Robot Platform	Dataset/ Environment	Evaluation Metrics	Main Findings	Limitations	Relevance to ARC-RVI
				with chance constraints for collision avoidance						
Visibility-Risk Navigation 6	Ghaleb et al., 2026, Machines	10.3390/machines14090980	Conflict between visibility-dependent tasks and physical navigation progress due to moving obstacles	VISTA-CBF+ELR framework combining visibility-risk planning, target-bearing control, ELR supervisor, CBF quadratic program	Mobile robots (locked factory simulations and Gazebo trials)	Factory simulations and Gazebo	Visibility maintenance, collision avoidance, navigation progress	CBF filtering maintains hard collision constraints while relaxing field-of-view requirements	Simulation-only; no physical robot validation	Medium: Demonstrates visibility-aware navigation but lacks physical validation
OmniPlanner 7	Zacharia et al., 2026, arXiv	10.48550/arxiv.2603.04284	Domain-specific, fragmented path planning across aerial, ground, underwater robots	Domain-agnostic bifurcated local-global planning kernel with platform abstraction layer and modular objective functions	Aerial robots (AR-1, AR-2), quadruped (ANYmal/GR-1), underwater ROV (UR-1), marsupial ground-aerial team	Underground mine, university campus, forest, ballast water tank, submarine bunker	Coverage completeness, execution time, path length	Universal planning framework applicable across robot morphologies	Limited details on inspection-specific quality metrics	Medium: Demonstrates domain-agnostic planning but lacks inspection focus
NEAT-based 3D Planning 8	Zhang et al., 2025	10.1051/matecconf/202541305002/pdf	Coverage completeness and path executability for mobile robotic arm NDT	NEAT algorithm with offline IK checking; point cloud graph structure; optimizes coverage, path smoothness, path length	Mobile robotic arm (Husky chassis)	ROS + Rviz simulation across different surface geometries	Coverage completeness, path smoothness, path length	NEAT-based planning with offline IK validation ensures executable paths	Simulation-only; no physical robot validation	Medium: Demonstrates coverage planning with IK validation but lacks physical validation
Directional Sensor Planning 9	Gemerek et al., 2022, IEEE Trans. Robotics	10.1109/TRO.2022.3180628	Path planning for mobile directional sensor with line-of-sight visibility and bounded field-of-view constraints	Novel approach accounting for visibility, field-of-view, obstacle avoidance, robot geometry, kinodynamic constraints	Camera-equipped drone	Real-world experiments and computational complexity analysis	Complete target visibility, distance traveled	Guarantees complete target visibility while minimizing distance	Limited to UAVs; no humanoid validation	Medium: Demonstrates visibility-aware planning but limited to UAVs
Joint View Planning 10	Zhang et al., 2026, IEEE RA-L	10.1109/lra.2025.3641156	Redundant viewpoints and decoupled viewpoint/path planning reduce inspection efficiency	Hybrid clustering and projection-covering optimization; multi-objective evolutionary framework for Set Covering TSP	Drone for building facade inspection	Large-scale building facade inspection	Coverage completeness, flight path length	Joint optimization of viewpoint selection and path planning improves efficiency	Limited to UAVs; no humanoid validation	Medium: Demonstrates joint optimization but limited to UAVs

Summary: Viewpoint and motion planning research demonstrates strong progress in coverage optimization, kinematic feasibility, and collision avoidance, but most methods target manipulators or UAVs. Humanoid-specific planning with whole-body kinematics and bipedal stability constraints remains underexplored. Physical validation is limited, and integration with image quality metrics is rare.

A.4 RVI-04: Active Visual Coverage — Literature Evidence Matrix

Paper	Authors & Venue	DOI	Research Problem	Core Methodology	Robot Platform	Dataset/ Environment	Evaluation Metrics	Main Findings	Limitations	Relevance to ARC-RVI
Cluttered Pipe Networks 1	Bettles et al., 2026	10.48550/arxiv.2603.16471	Autonomous 3D reconstruction and surface coverage of confined, cluttered pipe networks	Coverage-first NBV planner combining visual and coverage information gains; stochastic vector-field inequalities with chance constraints	UF7 manipulator on AgileX Tracer mobile platform	Confined space (1.5m cube with varying pipe configurations)	Coverage completeness, collision avoidance	Coverage-first planning with probabilistic collision avoidance enables reliable pipe network inspection	Limited to mobile manipulator; no humanoid validation	High: Demonstrates coverage-first NBV planning
Depth-Variance NBV 2	Desikan et al., 2026	10.1109/case69030.2026.11704184	Autonomous 3D reconstruction with coverage	Incremental 3D Gaussian splatting with depth-variance	UAV with RGB-D camera	PX4 drone in Gazebo simulation (30 trials)	Surface coverage, reconstruction quality	Depth-variance guidance prevents	Simulation-only; no physical robot	High: Demonstrates uncertainty-aware NBV

Paper	Authors & Venue	DOI	Research Problem	Core Methodology	Robot Platform	Dataset/ Environment	Evaluation Metrics	Main Findings	Limitations	Relevance to ARC-RVI
			metric saturation from hallucinated renders	acquisition function; measures per-pixel geometric uncertainty via alpha-blended depth second moments				coverage metric saturation and improves reconstruction	validation	planning
Motion-Uncertainty-Aware NBV 3	Li et al., 2026, arXiv	10.48550/arxiv.2605.17593	Active 3D reconstruction of unknown rigid object undergoing planar motion	Evaluates candidate viewpoints over plausible future object states induced by motion and measurement uncertainty; fixed-lag Gaussian Process smoother	Mobile robot	Simulation and real-world experiments	Reconstruction completeness	Motion-uncertainty-aware planning improves reconstruction of moving objects	Limited details on robot platform	High: Demonstrates uncertainty-aware NBV for moving objects
Adaptive UAV Planning 4	Jin et al., 2025, arXiv	10.48550/arxiv.2504.09294	UAV surface inspection in partially unknown indoor environments with unforeseen obstacles	Hierarchical framework: segment-based global viewpoint planner + local adaptive planner; region growing, LKH solver, B-spline trajectory, MPC-based obstacle avoidance	UAV with RGB-D camera and LiDAR	Simulation (two structured indoor environments) and physical flight test (tunnel)	Coverage completeness, execution time, obstacle avoidance	Adaptive planning enables reliable inspection in partially unknown environments	Limited to UAVs; no humanoid validation	High: Demonstrates adaptive planning under partial observability
Optimized AUV Exploration 5	Vidal et al., 2018, ICRA	10.1109/ICRA.2018.8460919	Automated exploration and inspection of unexplored underwater structures without prior maps	Combines view planning and frontier-based approaches; quadtrees for efficient viewpoint generation; hit-and-miss sonar noise filtering	Sparus II AUV	Simulation and real-world underwater experiments	Coverage completeness, exploration efficiency	Frontier-based exploration with view planning enables reliable underwater inspection	Limited to AUVs; no humanoid validation	Medium: Demonstrates frontier-based exploration but limited to AUVs
Active Perception Network 6	Vutetakis et al., 2023, arXiv	10.48550/arxiv.2309.11695	Online sensor-based coverage planning and visual surface coverage of unknown environments	Active Perception Network (hierarchical topological graph); Differential Regulation for incremental updates; hierarchical planner for non-myopic global exploration	AscTec Firefly MAV with stereo depth sensor	ROS-based Gazebo simulations	Coverage completeness, computational efficiency	Non-myopic planning via hierarchical topological graphs improves coverage efficiency	Simulation-only; no physical robot validation	Medium: Demonstrates non-myopic planning but lacks physical validation
Surface Edge Explorer 7	Border et al., 2018, ICRA	10.1109/ICRA.2018.8461098	NBV planning for surveying 3D scenes to improve scene models autonomously	Scene-model-free NBV planning using density representation; detects and explores observed surface boundaries	Not specified	Simulation	Surface coverage, computation time	Density-based NBV planning achieves better coverage in lower computation time	Limited details on robot platform and physical validation	Medium: Demonstrates density-based NBV planning
RobOralScan 8	Lee et al., 2026	—	Full-arch and long-span intraoral scanning for robotic dental reconstruction	Reinforcement learning-based pipeline with geometric memory-based observation space and tooth-wise coverage learning	Robot-scanner setup	Simulation and real-world experiments (8 of 10 evaluation episodes successful)	Scan completion, coverage completeness	RL-based active scanning enables reliable dental reconstruction	Limited to dental scanning; domain-specific	Low: Domain-specific (dental scanning); limited relevance to general inspection
Robot Active Neural Sensing 9	2022	10.48550/arxiv.2208.11079	Active sensing and scene reconstruction	Dual approach: 3DCNN-based ScoreNet +	6-DoF UR5e manipulator with Intel	Nvidia Isaac Gym simulation and 10 real-world	Coverage completeness, number of viewpoints	Learning-based NBV planning with shape	Limited to manipulator; no humanoid	High: Demonstrates learning-based NBV

Paper	Authors & Venue	DOI	Research Problem	Core Methodology	Robot Platform	Dataset/ Environment	Evaluation Metrics	Main Findings	Limitations	Relevance to ARC-RVI
			in unknown, cluttered, narrow environments	bilevel MPC or Transformer-based VPFormer; object-wise shape completion	RealSense D435i	cabinet settings with YCB objects		completion minimizes required viewpoints	validation	with shape completion
Prediction-Based NBV 10	Wang et al., 2025, ISPRS Annals	10.5194/isprs-annals-x-2-w2-2025-207-2025	High-quality 3D reconstruction with sufficient scene coverage during online exploration	MACARONS NBV method using monocular RGB sensor; generates optimized global exploration trajectories maximizing surface coverage gain	UAV	Two open-access 3D bridge models (Sketchfab)	Coverage completeness, reconstruction quality	Prediction-based NBV achieves high-quality reconstruction with efficient coverage	Simulation-only; no physical robot validation	Medium: Demonstrates prediction-based NBV but lacks physical validation

Summary: Active visual coverage research demonstrates strong progress in information-theoretic planning, learning-based methods, and application-specific adaptation. However, quality-aware planning integrating image quality metrics is rare, humanoid platform evaluation is limited, and sim-to-real validation is often lacking. Most methods optimize coverage efficiency without considering acquisition repeatability or documentation quality.

A.5 RVI-05: Repeatable Visual Acquisition — Literature Evidence Matrix

Paper	Authors & Venue	DOI	Research Problem	Core Methodology	Robot Platform	Dataset/ Environment	Evaluation Metrics	Main Findings	Limitations	Relevance to ARC-RVI
Active Pose Relocalization 2	2023	10.1109/tie.2022.3186368	Active pose relocalization for intelligent substation inspection robot	Active visual servo relocalization for omnidirectional inspection robots	Substation inspection robot	Substation environments	Relocalization accuracy	Active pose relocalization improves inspection consistency	Limited details on acquisition geometry repeatability	Medium: Demonstrates pose relocalization but limited to robot pose
Vision-Based Error Compensation 3	Olzhayev et al.	—	Vision-based error compensation for accurate robotic positioning in automated marking	Coordinate transformation framework for vision-guided positioning	Robotic marking system	Automated vehicle body marking	Positioning accuracy	Vision-based error compensation improves positioning accuracy	Limited details on acquisition geometry repeatability	Medium: Demonstrates positioning accuracy but limited to marking tasks
Active Visual Servo Relocalization 4	Zheng et al., 2026	10.1109/ispp69262.2026.11542998	Active visual servo relocalization for omnidirectional substation inspection robots	Active visual servo relocalization	Omnidirectional substation inspection robot	Substation environments	Relocalization accuracy	Active visual servo relocalization improves inspection consistency	Limited details on acquisition geometry repeatability	Medium: Demonstrates relocalization but limited to robot pose
Photogrammetric Multi-Camera Calibration 7	Ribeiro et al., 2019, ISPA	10.1109/ISPA.2019.8868928	Photogrammetric multi-camera calibration using industrial programmable robotic arm	Photogrammetric calibration using robotic arm	Industrial programmable robotic arm	Calibration environments	Calibration accuracy	Precise extrinsic parameter estimation using robotic arm	Limited details on acquisition geometry repeatability	Medium: Demonstrates calibration accuracy but limited to camera calibration
Sensor-Based Robotic Line Scan 8	Huo et al., 2022, IEEE Sensors Journal	10.1109/jsen.2021.3132428	Sensor-based robotic line scan system with adaptive ROI for defect detection over convex free-form specular surfaces	Adaptive ROI line scan system	Robotic line scan system	Convex free-form specular surfaces	Defect detection accuracy	Adaptive ROI improves defect detection on specular surfaces	Limited details on acquisition geometry repeatability	Low: Focuses on defect detection, not acquisition repeatability
Robot Arm Repeated Positioning 12	Zhang et al., 2018, ICSPCC	10.1109/ICSPCC.2018.8567779	Robot arm repeated positioning accuracy measurement	Nonlinear local feature matching for positioning accuracy measurement	Robot arm	Positioning accuracy measurement	Positioning accuracy, repeatability	Nonlinear local feature matching enables accurate positioning measurement	Measures robot pose repeatability, not acquisition geometry	Medium: Demonstrates positioning repeatability measurement
Robust Visual Teach-and-Repeat 13	Mahdavian et al., 2021	—	Robust visual teach-and-repeat for UGVs using 3D semantic maps	3D semantic maps for teach-and-repeat navigation	UGV	Outdoor environments	Navigation repeatability	3D semantic maps enable robust teach-and-repeat navigation	Limited details on acquisition geometry repeatability	Medium: Demonstrates navigation repeatability but limited to localization
Hand-Eye Calibration 17	Braga et al., 2026	10.1109/codit70676.2026.11630871	Comparison of automatic hand-eye calibration methods for vision-guided collaborative robots	Compares automatic hand-eye calibration methods	Collaborative robots	Calibration environments	Calibration accuracy	Systematic comparison of hand-eye calibration methods	Limited details on acquisition geometry repeatability	Medium: Demonstrates calibration methods but limited to camera calibration

Paper	Authors & Venue	DOI	Research Problem	Core Methodology	Robot Platform	Dataset/ Environment	Evaluation Metrics	Main Findings	Limitations	Relevance to ARC-RVI
Hybrid Visual Servoing 18	Danesh et al., 2025	10.2139/ssrn.5175232	Hybrid visual servoing of tendon-driven continuum robots	Combines image-based and position-based visual servoing	Tendon-driven continuum robot	Visual servoing tasks	Servoing accuracy	Hybrid visual servoing improves control accuracy	Limited details on acquisition geometry repeatability	Medium: Demonstrates visual servoing but limited to control accuracy
Mobile Collaborative Robot Accuracy 24	Hifi et al., 2025, Meas. Sci. Technol.	10.1088/1361-6501/adf3d4	Positional uncertainty of mobile collaborative robots in high-value manufacturing inspection	Precision laser tracker evaluation of integrated feature mapping, varying speeds, translation orientations	KUKA KMR iiwa mobile manipulator	High-value manufacturing environments	Positional accuracy, repeatability	Quantitative evaluation reveals positional uncertainty factors	Measures robot pose repeatability, not acquisition geometry	High: Demonstrates quantitative repeatability evaluation but limited to robot pose

Summary: Repeatable visual acquisition research focuses primarily on robot pose repeatability, camera calibration, and visual servoing, with limited attention to acquisition geometry repeatability, image quality consistency, or documentation fidelity. Humanoid platform evaluation is rare, and standardized benchmarks for acquisition repeatability are lacking.

A.6 RVI-06: Acquisition Quality Control — Literature Evidence Matrix

Paper	Authors & Venue	DOI	Research Problem	Core Methodology	Robot Platform	Dataset/ Environment	Evaluation Metrics	Main Findings	Limitations	Relevance to ARC-RVI
Auto-Exposure Algorithm 3	Begin et al., 2022, Sensors	10.3390/s22030835	Enhancing mobile robot visual localization in challenging light conditions	Maximizes gradient information; leverages photometric response function; incorporates saturation feedback; balances gain and exposure time using optical flow	Stereo cameras on custom three-axis motion table	Real-world experiments with ORB-SLAM3	Gradient information, localization accuracy	Dynamic exposure control improves visual odometry performance	Limited to localization; does not address inspection-specific quality metrics	High: Demonstrates adaptive exposure control for robust perception
Automated Camera Exposure 4	Zhang et al., 2022, Sensors	10.3390/s22249641	Acquiring high-quality speckle image pairs in stereo-DIC without physical touch under varying ambient light	Determines optimal exposure times and ideal average grayscales; uses maximum mean intensity gradient for reference images; adaptive exposure iteration for deformed images	Stereo-DIC system with two FLIR-GS3-U3 cameras	Real-world experiments including high-temperature testing	Average grayscale, intensity gradient, measurement accuracy	Automated exposure control enables accurate stereo-DIC measurement	Limited to stereo-DIC; does not address general inspection quality	Medium: Demonstrates automated exposure control but domain-specific
Predictive Exposure Control 6	Deng et al., 2024, Robot. Comput.-Integr. Manuf.	10.1016/j.rcim.2023.102619	Poor lighting performance in vision-based robotic disassembly	ROI extraction; entropy-based quality assessment; exposure-entropy prediction model	Robot vision system	Screw-removal case study in EV battery disassembly	Image quality, exposure time	Predictive exposure control improves vision-based disassembly	Limited to disassembly tasks; does not address general inspection	Medium: Demonstrates predictive exposure control but domain-specific
Low Quality Image Detection 7	Perić et al., 2022, Appl. Sci.	10.3390/app12063041	Automatic detection and rejection of low-quality, blurred images in railway obstacle detection	Improved edge detection with Gaussian noise reduction; automatic day/night threshold calculation	Cameras mounted in front of train	Railway dataset (2,180 images)	Blur detection accuracy	Automatic blur detection improves training dataset quality	Limited to railway obstacle detection; does not address general inspection	Low: Domain-specific (railway); limited relevance to general inspection
Noise-Aware Exposure Control 8	Shin et al., 2019, IROS	10.1109/IROS40897.2019.8968590	Robust robot vision through camera exposure control	Noise-aware quality metric combining gradient, entropy, noise; real-time automatic exposure and gain control using Nelder-Mead method	Not specified	Large set of experimental results	Image quality, gradient, entropy, noise	Noise-aware exposure control improves visual perception robustness	Limited details on robot platform and physical validation	High: Demonstrates noise-aware quality assessment

Paper	Authors & Venue	DOI	Research Problem	Core Methodology	Robot Platform	Dataset/ Environment	Evaluation Metrics	Main Findings	Limitations	Relevance to ARC-RVI
Learning-Based Exposure Control 9	Lee et al., 2024, arXiv	10.48550/arxiv.2404.01636	Adjusting camera exposure in arbitrary and dynamic lighting conditions	Deep reinforcement learning for rapid exposure control; flickering-aware reward; static-to-dynamic lighting curriculum; domain randomization	Simplified training ground with domain randomization	Real-time processing (1 ms, five steps)	Exposure level, feature extraction, object detection	RL-based exposure control achieves rapid adjustment in dynamic lighting	Limited details on physical robot validation	High: Demonstrates learning-based exposure control
Active Observation 10	Hu et al., 2026	10.1109/case69030.2026.11704293	Approach-phase occlusion disrupting closed-loop control during robotic grasping	Active observation framework using dual-arm platform; occlusion-triggered exploration module	Dual 6-DoF Piper platform	Real-world experiments	Grasp success rate, occlusion recovery	Active observation recovers shared target-gripper view under occlusion	Limited to grasping; does not address general inspection quality	Medium: Demonstrates active observation for occlusion recovery
Quality-Aware Re-Observation 12	Zhao et al., 2026, Appl. Sci.	10.3390/app16189389	Unreliable local stereo depth despite object visibility in constrained eye-in-hand stereo manipulators	Elevates ROI-level stereo reliability to admission condition; online proxy strategy based on Jproxy; generates four candidate viewpoints filtered by constraints	Four-DoF eye-in-hand stereo manipulator	Real-world experiments	Stereo reliability, re-observation frequency	Quality-aware re-observation improves stereo depth reliability	Limited to stereo manipulators; does not address general inspection	High: Demonstrates quality-aware re-observation
Enhanced View Planning 13	Li et al., 2025, arXiv	10.48550/arxiv.2503.10334	Occlusion challenges in automated robotic harvesting	Imitation learning-based viewpoint planning using ACT algorithm; predicts 6-DoF camera motion actions from RGB-D images	6-DoF UR5e robotic arm with Intel RealSense D435	Simulation (ROS + Gazebo) and real-world laboratory	Occlusion recovery, viewpoint quality	Imitation learning enables effective occlusion-aware viewpoint planning	Limited to agricultural harvesting; does not address general inspection	Medium: Demonstrates learning-based occlusion-aware planning
Proactive Camera Control 15	Kim et al., 2020, IEEE Trans. Robotics	10.1109/TRO.2020.2985597	Vision algorithm failure under radical illumination variance	Bayesian optimization for exposure time and gain control; image metric combining gradient and SNR; synthesizes images using camera response function	Real-world experiments (indoor, outdoor, transient environments)	Light-flickering indoor, outdoor near sunset, indoor-outdoor transient	Image quality, frame rate (20-40 Hz)	Proactive camera control maintains robust perception under illumination variance	Limited details on robot platform	High: Demonstrates proactive camera control for robust perception

Summary: Acquisition quality control research demonstrates strong progress in adaptive exposure control, noise-aware quality assessment, and motion blur detection. However, most methods operate in isolation from viewpoint planning and coverage optimization. Integration of quality metrics into active perception and NBV planning is rare, and humanoid platform validation is limited.

A.7 RVI-07: Simulation-to-Reality Transfer — Literature Evidence Matrix

Paper	Authors & Venue	DOI	Research Problem	Core Methodology	Robot Platform	Dataset/ Environment	Evaluation Metrics	Main Findings	Limitations	Relevance to ARC-RVI
da Vinci Sim2Real 1	Haiderbhai et al., 2022, IROS	10.1109/IROS47612.2022.9981573	Sim-to-real transfer gap in autonomous surgical robotics	ROS-based sim-to-real pipeline with Unity3D dVRK simulation; domain randomization across camera, lighting, physics variables	Unity3D da Vinci Research Kit simulation; OAK-1 and dVSS Endoscope cameras	Cube pushing task	Success rate across simulation, OAK-1, dVSS Endoscope	Domain randomization enables sim-to-real transfer for surgical manipulation	Limited to surgical robotics; does not validate visibility models	Medium: Demonstrates domain randomization but limited to manipulation

Paper	Authors & Venue	DOI	Research Problem	Core Methodology	Robot Platform	Dataset/ Environment	Evaluation Metrics	Main Findings	Limitations	Relevance to ARC-RVI
Robust Visual Sim2Real 2	García et al., 2023, IROS	10.1109/iros55552.2023.10342471	Visual sim-to-real domain gap in robotic manipulation	Benchmarks visual domain randomization methods; offline proxy task (cube localization) for parameter selection	Real robot experiments	Challenging manipulation tasks	Success rate (93% average)	Offline parameter selection enables effective sim-to-real transfer	Limited to manipulation; does not validate visibility models	Medium: Demonstrates visual domain randomization but limited to manipulation
Pose Estimation Sim2Real 3	Lu et al., 2020	10.1109/LRA.2022.3151981	Detecting user-defined, marker-less keypoints on robot manipulators	Autonomous optimization for optimal keypoint locations; trains DNNs using synthetic data; domain randomization for sim-to-real transfer	Robot manipulators	Simulation and real robotic applications (calibration, tracking, pose estimation)	Pose estimation accuracy	Domain randomization enables zero-shot sim-to-real transfer for pose estimation	Limited to pose estimation; does not validate visibility models	Low: Focuses on pose estimation, not inspection
Visual Dexterity 4	Chen et al., 2022, arXiv	10.48550/arXiv.2211.11744	Dynamic in-hand object reorientation without object-specific pose estimators	Two-stage teacher-student RL framework; synthetic and rendered point clouds; dynamics identification via black-box parallel simulation; domain randomization	D'Claw manipulator (three-fingered, nine DoF) and modified four-fingered version (twelve DoF)	Isaac Gym simulation and real-world experiments with 3D-printed and household objects	Reorientation success rate	Sim-to-real transfer enables in-hand manipulation without pose estimators	Limited to manipulation; does not validate visibility models	Low: Focuses on manipulation, not inspection
DiffuDepGrasp 5	Zhou et al., 2025, arXiv	10.48550/arxiv.2511.12912	Sim2real gap caused by sensor artifacts in real depth maps	DiffuDepGrasp framework with diffusion depth generator; temporal geometric prior diffusion depth module; noise grafting module	Simulation-exclusive training with zero-shot transfer to physical robots	12-object grasping setup	Grasp success rate (95.7% average)	Diffusion-based depth noise modeling enables zero-shot sim-to-real transfer	Limited to grasping; does not validate visibility models	Low: Focuses on grasping, not inspection
Vid2Param 6	Asenov et al., 2019	—	Online system identification from video for robotics applications	Vid2Param extends Variational RNNs with physics-based dynamics model conformity loss	PR2 robot with velocity-constrained arm and standard camera	Simulation and real-world PR2 experiments (ball interception)	Trajectory prediction accuracy, physical parameter estimation	Online system identification from video enables sim-to-real transfer	Limited to trajectory prediction; does not validate visibility models	Low: Focuses on system identification, not inspection
Domain Randomization for Pose 7	Ren et al., 2019, ICRA	10.1109/ICRA.2019.8794126	Improving accuracy of domain-randomization-based pose estimation	Active perception with domain randomization; trains neural network in simulation with high texture/lighting variations	Robotic manipulation tasks (robot holding object, moving reference objects, moving camera)	Simulation and real-world transfer	Pose estimation accuracy	Active perception with domain randomization improves pose estimation	Limited to pose estimation; does not validate visibility models	Low: Focuses on pose estimation, not inspection
Sim2Real Dynamic Grasping 8	Luo et al., 2026	10.1109/icmtim69588.2026.11525620	Insufficient policy generalization induced by Sim2Real gap in dynamic grasping	Multimodal end-to-end method with domain randomization; fuses RGB images, depth maps, robot body states	Multimodal perception network	Simulation (nearly 100% success at 0.15-0.40 m/s) and physical experiments (95.0% static, 90.0% low-speed dynamic)	Grasp success rate	Comprehensive domain randomization enables zero-shot sim-to-real transfer	Limited to grasping; does not validate visibility models	Low: Focuses on grasping, not inspection
Humanoid Grasping Sim2Real 9	Gaede et al., 2022, LNCS	10.1007/978-3-031-15919-0_29	Domain randomization effectiveness for real-world performance without fine-tuning	Replicates reach-to-grasp experiment with NICO humanoid robot; autonomous training data creation; supervised learning with end-to-end CNN	NICO humanoid robot	Simulation and real-world reach-to-grasp experiments	Grasp success rate	Domain randomization increases real-world performance without fine-tuning	Limited to grasping; does not validate visibility models	Medium: Demonstrates humanoid sim-to-real transfer but limited to grasping
Viral 14	He et al., 2025, arXiv	10.48550/arxiv.2511.15200	Lack of autonomous loco-	Teacher-student design with privileged	Unitree G1 humanoid robot	Large-scale simulation (up to 64	Loco-manipulation success rate	Large-scale visual domain randomization	Limited to loco-manipulation;	High: Demonstrates humanoid

Paper	Authors & Venue	DOI	Research Problem	Core Methodology	Robot Platform	Dataset/ Environment	Evaluation Metrics	Main Findings	Limitations	Relevance to ARC-RVI
			manipulation skills for real-world humanoid robot deployment	RL teacher and vision-based student; large-scale visual domain randomization across lighting, materials, camera parameters, image quality, sensor delays		GPUs) and real-world deployment	(up to 54 cycles)	enables zero-shot humanoid loco-manipulation	does not validate visibility models	sim-to-real transfer on Unitree G1

Summary: Sim-to-real transfer research demonstrates strong progress in domain randomization, domain adaptation, and real-to-sim calibration for manipulation tasks. However, most studies focus on manipulation success rates rather than visibility model accuracy, coverage metric fidelity, or inspection-specific requirements. Humanoid platform validation is rare, and systematic validation of simulated visibility and coverage against physical measurements is lacking.

A.8 RVI-08: End-to-End Autonomous Inspection — Literature Evidence Matrix

Paper	Authors & Venue	DOI	Research Problem	Core Methodology	Robot Platform	Dataset/ Environment	Evaluation Metrics	Main Findings	Limitations	Relevance to ARC-RVI
RoboInspect 1	Wang et al., 2026, Comput.-Aided Civ. Infrastruct. Eng.	10.1016/j.cacaie.2026.100007	Unexplored potential of humanoid robots in indoor structural health monitoring	End-to-end autonomous inspection framework with RL-based locomotion, autonomous navigation, vision-based crack detection	Full humanoid robot	Two real-world case studies in university building	Navigation reliability, crack detection accuracy	Reliable autonomous navigation and crack detection in controlled conditions	No quantitative comparison against human inspectors	High: Demonstrates end-to-end humanoid inspection framework
Cyber-Physical Inspection 2	Hotchandani et al., 2026, Sci. Rep.	10.1038/s41598-026-56206-0	Precise, scalable, traceable, compliant cyber-physical inspection in regulated manufacturing	Pilot architecture integrating collaborative robotics, structured-light scanning, industrial IoT; Gauge R&R studies	Collaborative robotics with in-house spray setup; Zeiss ScanBox	Controlled production environment	Repeatability, cycle time	Standards-aligned architecture enables traceable inspection	Limited details on autonomous vs. manual comparison	Medium: Demonstrates system integration but lacks autonomous comparison
Comparative Analysis 7	Aust et al., 2022, Appl. Sci.	10.3390/app12042250	Aircraft maintenance inspection prone to human error	Compares human operators against image processing, AI, 3D scanning across piece-part, borescope, visual-tactile inspection	Image processing, AI, 3D scanning (GOM Atos Q scanner)	Benchmark blade images, borescope stills, physical blade parts	Inspection accuracy, consistency, time, SWOT, weighted factor analysis	Technology advantages in consistency; human superiority in complex defect interpretation	Limited to aero engine blade inspection	High: Demonstrates quantitative human-technology comparison
Human vs. Robot Benchmarking 22	Xia et al., 2022, J. Constr. Eng. Manag.	10.1061/(asce)co.1943-7862.0002322	Deeper insights and baseline models for robot-assisted emergency structural inspection	Benchmarks human vs. robot performance in simulated emergency structural inspection	Simulation	Simulated emergency structural inspection	Total identified damage, inspection time, route patterns	Robots excel in systematic coverage; humans demonstrate superior adaptability	Simulation-only; no physical robot validation	Medium: Demonstrates human-robot comparison but lacks physical validation
Mobile Collaborative Robot Accuracy 21	Hifi et al., 2025, Meas. Sci. Technol.	10.1088/1361-6501/adf3d4	Positional uncertainty of mobile collaborative robots in high-value manufacturing inspection	Precision laser tracker evaluation of integrated feature mapping, varying speeds, translation orientations	KUKA KMR iiwa mobile manipulator	High-value manufacturing environments	Positional accuracy, repeatability	Quantitative evaluation reveals positional uncertainty factors	No comparison against human or teleoperated inspection	Medium: Demonstrates quantitative accuracy evaluation but lacks comparison
Bridge Inspection Comparison 26	Kim et al., 2022, Drones	10.3390/drones6110355	Integration and automation of bridge inspection	Compares UAS-based automatic damage detection against traditional ladder truck inspection	Customized quad-rotor UAV with Pixhawk 2, Intel i7 NUC, Velodyne Puck lite LiDAR, DSLR camera	D Bridge in Gangwon province, Korea	Damage detection accuracy, inspection time	UAS-based inspection achieves comparable accuracy with reduced time and risk	Limited to bridge inspection; does not address general inspection	Medium: Demonstrates UAS-human comparison but domain-specific

Paper	Authors & Venue	DOI	Research Problem	Core Methodology	Robot Platform	Dataset/ Environment	Evaluation Metrics	Main Findings	Limitations	Relevance to ARC-RVI
ANYmal Offshore Inspection 20	Gehring et al., 2021, Field Serv. Robot.	10.1007/978-981-15-9460-1_18	Challenging environment of offshore HVDC platforms where mobile robots have not been used	Deploys ANYmal quadrupedal robot for field tests on offshore platform	ANYmal quadrupedal robot with visual, thermal, acoustic sensors	Offshore HVDC platform in North Sea	Navigation, sensor payloads, automated instrument reading, coverage	Successful field deployment demonstrates feasibility of robotic offshore inspection	No quantitative comparison against human inspectors	High: Demonstrates real-world field deployment
Semi-Autonomous Teleoperation 4	Marquette et al., 2024	10.48550/arxiv.2412.10973	Teleoperation challenges in confined-space aircraft wing inspection	Novel crane robot with differential flatness-based control; semi-autonomous control modulates operator inputs	Crane robot traversing stringer channels with suspended camera	Real-world autonomous and teleoperated user experiments (12 participants) and numerical simulations	Collision avoidance, oscillation reduction	Semi-autonomous control eliminates collisions and reduces oscillations	Limited to teleoperation; does not compare fully autonomous inspection	Medium: Demonstrates semi-autonomous teleoperation but lacks autonomous comparison
Human-Autonomy Teaming 11	Xia et al.	10.1061/jccee5.cpeng-5756	High learning and operational burdens of traditional ROV joystick controls	VR-based sensory augmentation and body motion teleoperation; interactive UI for seamless engagement and dynamic switching	Remotely operated vehicle (ROV)	Subsea inspection tasks	Inspection performance (human control, full autonomy, teaming)	Human-autonomy teaming improves inspection performance	Limited to teleoperation; does not compare fully autonomous inspection	Medium: Demonstrates human-autonomy teaming but lacks autonomous comparison
Semi-Autonomous Mobile Robot Control 12	Seppänen et al., 2022, J. Intell. Robot. Syst.	10.1007/s10846-022-01711-3	Large delay fluctuation in communication signals during mobile robot teleoperation	Compares fully manual, autonomous, delayed manual, semi-autonomous strategies; RGB-D image-based artificial potential field	3-wheeled differential drive mobile robot with Intel RealSense D455	Mock-up environment mimicking particle accelerator tunnel	Navigation performance under delay	Semi-autonomous strategies improve performance under large delay fluctuation	Limited to teleoperation; does not compare fully autonomous inspection	Medium: Demonstrates semi-autonomous control but lacks autonomous comparison

Summary: End-to-end autonomous inspection research demonstrates system integration and field deployment, but quantitative comparisons of autonomous vs. manual inspection are scarce. Most studies compare teleoperation or semi-autonomous control rather than fully autonomous systems. Humanoid platform evaluation is rare, and controlled experimental designs with standardized tasks and statistical analysis are limited.

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